Algorithm 1 Per-QAM margin search and refinement

Input: per-tone SNRs $\{\gamma_k\}$; base thresholds $\{T_M\}$.
Output: per-tone QAM allocation $\{M_k\}$.
$\Delta = 6554/2^{16}$ dB; $P_{b,\text{target}} = 10^{-4}$.

$\mu \leftarrow \mathbf{0}$; $P_{\text{acc}} \leftarrow P_{b,\text{target}} - 2^{-20}$
$(P, \cdot, \{\bar{p}_M\}) \leftarrow \text{Scan}(\mu)$
while $P > P_{\text{acc}}$ **do**
    $M_w \leftarrow \arg\max_{M \text{ active}} \bar{p}_M$ (ties: larger M)
    $\mu_{M_w} \leftarrow \mu_{M_w} + \Delta$
    $(P, \cdot, \{\bar{p}_M\}) \leftarrow \text{Scan}(\mu)$
repeat
    $(\mu^*, B^*) \leftarrow (\mu, -1)$
    for $M \in (4, 16, 64, 256)$ **with** $\mu_M \geq \Delta$ **do**
        $\mu' \leftarrow \mu$; $\mu'_M \leftarrow \mu_M - \Delta$
        $(P', B', \cdot) \leftarrow \text{Scan}(\mu')$
        if $P' \leq P_{\text{acc}}$ **and** $B' > B^*$ **then**
            $(\mu^*, B^*) \leftarrow (\mu', B')$
    $\mu \leftarrow \mu^*$
until $B^* = -1$
return FinalScan(μ)

Scan(μ) selects the highest QAM order satisfying $\gamma_{k,\text{dB}} \geq T_M + \mu_M$ (otherwise the tone is unused), then returns bit-weighted estimated BER P, total loaded bits B, and mean BER $\bar{p}_M$ for each nonempty QAM group. Set $P = 0$ when $B = 0$. FinalScan streams the retained allocation.

Normal search is shown; RTL watchdog exits are omitted.